# Satellite Communication Modes

| **Mode** | **Telecommand (BFSK)** *437.100 MHz* | **Telemetry (GFSK)** | **Data (QPSK )** | **VHF Beacon Signal (CW/OOK )** |
|---|---|---|---|---|
| Mode-1 (Default) | UHF-RX *437.100 MHz* | UHF-TX *437.325 MHz* | 0 | 0 |
| Mode-2 (General) | UHF-RX *437.100 MHz* | UHF-TX *437.325 MHz* | UHF-RXTX *436.500 / 435.800 MHz* | Beacon *145.84 MHz* |
| Mode-3 (ISL) | UHF-RX *437.100 MHz* | UHF-TX *437.325 MHz* | S-TXRX (ISL) *2245.000 MHz* | Beacon *145.84 MHz* |
| Mode -4 (HDD) | UHF-RX *437.100 MHz* | UHF-TX *437.325 MHz* | S-TX *2245.000 MHz* | Beacon *145.84 MHz* |
| Mode - 5 (Emergency) | UHF-RX *437.100 MHz* | VHF-TX *145.950 MHz* | 0 | Beacon *145.84 MHz* |

# Ground Station Communication Modes

| **Mode** | **Telecommand (BFSK)** | **Telemetry (GFSK)** | **DC (QPSK )** | **BEACON (CW/OOK )** 145.84 MHz |
|---|---|---|---|---|
| Mode-1 (Default) | UHF-TX *437.100 MHz* | UHF-RX 437.325 MHz | 0 | 0 |
| Mode 2 (General) | UHF-TX *437.100 MHz* | UHF-RX 437.325 MHz | UHF-TXRX *436.500/435.800 MHz* | VHF RX *145.84 MHz* |
| Mode 3 (ISL) | UHF-TX *437.100 MHz* | UHF-RX 437.325 MHz | 0 | VHF RX *145.84 MHz* |
| Mode -4 (HDD) | UHF-TX *437.100 MHz* | UHF-RX 437.325 MHz | S-RX 2245.000 MHz | VHF RX *145.84 MHz* |
| Mode - 5 (Emergency) | UHF-TX *437.100 MHz* | VHF-RX 145.950 MHz | 0 | VHF RX *145.84 MHz* |

# 0.Communication

In this section, specifications of the communication subsystem consistent with the expected attributes of detailed design are presented as follows:

- Detailed Communication system expression (Modulation selection, determining bitrate, coding, etc.)
- Architecture, block diagram and comprehensive hardware specification (Product tree)
- Simulation and presentation of radiation pattern, VSWR, axial ratio, and other specifications of TT&C antennas.
- Simulation and presentation of radiation pattern, VSWR, axial ratio, and other specifications of payload antennas.
- Final link budget calculations of TT&C in normal and reduced modes.
- Interfaces (structural, data, …).
- Subsystem detailed product tree and bill of materials (BOM)
- Definite budgets (cost, mass, power, spatial)